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Cognitive Sciences

Opinion

Language learning as ontogenetic adaptation

Manuel Bohn [1,2,*] and Marisa Casillas [3]

Language learning is a multi-threaded, multi-mechanism process. It is multi-threaded in that it emerges as a byproduct of addressing multiple goals while engaging in social interactions. It is multi-mechanism in that children integrate multiple information sources to infer what is meant and what to say next. These information sources include contextual and social cues, as well as cognitive mechanisms. Focusing on early word learning, this article reviews information sources, how children might sensitively adapt to them, and how we can model their integration using Bayesian inference over multiple probability distributions. We argue that, to advance our understanding of language learning, we must jointly study how children learn from multiple information sources across diverse developmental settings.

Highlights

Learning linguistic forms and processes is one of the multiple threads children engage in as they observe and participate in social interaction.

Recent work shows that the environment in which children learn language varies tremendously within and between communities, requiring flexibility on the side of the child to cope with such variability.

This flexibility is enabled by structural redundancy in the language learning process as children can rely on a broad range of information sources (i.e., contextual, social, and cognitive learning mechanisms).

The integration of different information sources has recently been described for referent resolution as a form of Bayesian inference; predictions from this account have been successfully tested against experimental data.

Children adapt to the information sources available in their language learning environment by fine-tuning their sensitivity to these sources.

Language learning in the real world

Children make meaning of language in the course of everyday interaction. They learn not just how to say things but when and why to say them. Put differently, they become not just language knowers but also language users, and they do so with little to no explicit language instruction [1]. Thus, linguistic representations are encountered—and then built—to accommodate use in real-world social interaction. And because everyday interactions vary across social, cultural, and institutional settings, we can expect the process of language development to vary too [2,3]. Yet consistencies in language learning across a range of domains highlight a core puzzle for language scientists [4–6]: how do human learners so flexibly adapt to environmental diversity in development? This review considers how such adaptability may arise through a multi-mechanism, multi-threaded process, driven by children's desire to effectively participate in interaction.

The argument we spell out in this article starts from the idea that, at any given moment, there are multiple—partly redundant—cognitive processes that can be applied to address the multi-threaded goals of spontaneous verbal interaction. Language environments differ in which processes are most effective because the information sources these processes operate on are differentially available across contexts. The two central premises of our proposal are that, when inferring the meaning of (multimodal) utterances: (i) children integrate multiple information sources, and (ii) over development, their cognitive architecture adapts to the distribution of information sources in their language learning environment.

In what follows, we review several processes involved in word learning, as well as variability in language learning environments. We then present a Bayesian modeling framework that allows us to conceptually link these two aspects of learning and thus formally conceptualize a theory of word learning that is rich in information sources and adaptive in using them to add new words. We then continue by reviewing existing evidence that children adapt their cognitive architecture to the characteristics of their language learning environment, ending with suggestions for future research that embraces and builds on this proposal.

[1]Institute of Psychology in Education, Leuphana University Lüneburg, Lüneburg, Germany
[2]Department of Comparative Cultural Psychology, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany
[3]Department of Comparative Human Development, University of Chicago, Chicago, IL, USA

*Correspondence:
manuel.bohn@leuphana.de (M. Bohn).

 https://doi.org/10.1016/j.tics.2025.12.005
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A wealth of information sources

We begin with an example: a parent and a 2-year-old child are in the kitchen. The parent is preparing food—a soup—and the child is playing nearby. To keep the child safely occupied, the parent occasionally hands the child a cooking utensil to play with. When the soup is ready, the parent turns toward the child, searches the pile of utensils, and says: 'Where did you hide that ladle?' We assume the child does not know the words 'hide' or 'ladle' (Figure 1).

When deciding what to do next, the child will engage in a multi-threaded process of solving problems in interaction and acquiring linguistic knowledge (e.g., learning the unknown words 'hide' and 'ladle'). The child might be, most urgently, trying to determine whether they need to make a response and, if so, what kind of response is needed. In designing a response, they can achieve two related outcomes: satisfying the request (i.e., handing the ladle over) and getting confirmatory evidence about linguistic forms (i.e., via parental acceptance of the object; Figure 1). At a broader level, their actions should align with their implicit and explicit ongoing commitments to joint action (e.g., exchanging items, staying in the kitchen) and, similarly, respect the relevant roles and social construction of the interaction (e.g., acquiescing to parental requests more often than peer requests). In this way, one response can progress multiple interactional and linguistic threads simultaneously. Language learning then comes as a byproduct of using language to coordinate moves within social interactions.

The process of language learning is multi-mechanism: the child has at their disposal a variety of cues (contextual, interactional, and social), cognitive biases, and learning mechanisms while engaging in social interaction. We will refer to these more generally as 'information sources'. Information sources orient children to relevant meanings in social interaction and thereby help children produce and understand language. As we argue in the section "Embracing variability

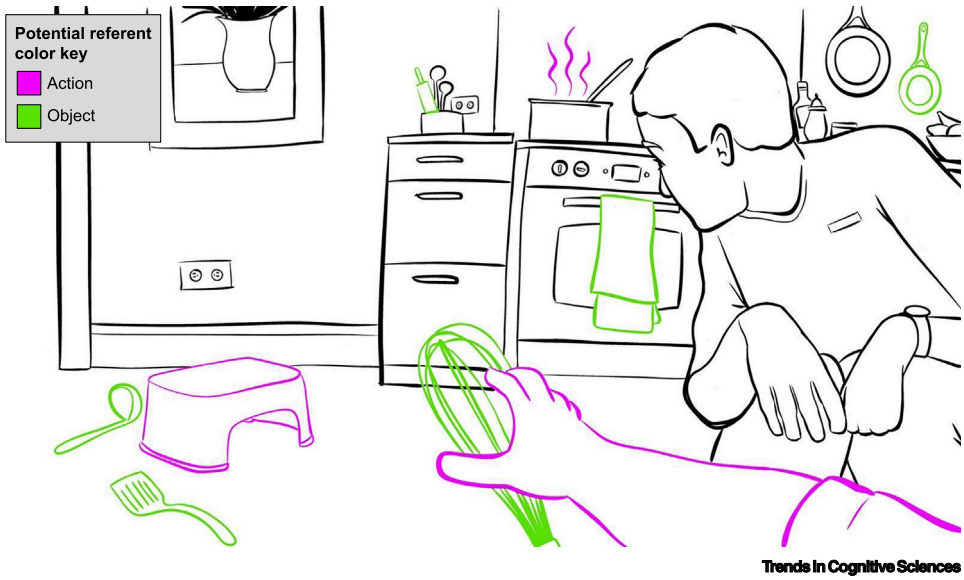


Figure 1. Learning language in real-time interaction. Schematic depiction of the example described in the text. The adult and child are engaged in an interaction involving multiple cooking utensils when the adult says, 'Where did you hide that ladle?' Learning the meanings of the unknown words 'hide' and 'ladle' is only one of multiple threads the child is engaged in (satisfying the request being another one). Potential referents for the noun are depicted in green, and potential referents for the verb are indicated in magenta. To infer the meanings of the unknown words, the child will rely on multiple information sources (contextual, social, and cognitive).

in language learning", this structural redundancy gives children the flexibility in adjusting to variable learning environments. We briefly review a number of such information sources relevant to our ladle-hiding example (Figure 1), here organized into three rough categories: contextual, social, and cognitive. This is not meant to be an exhaustive list of all potential information sources. For example, we did not discuss sound symbolism—the nonarbitrary mapping between words and their referents [7,8]. We also did not elaborate on the affordances of spontaneous interaction—the fact that utterances are embedded in ongoing conversation, allowing interactants to ask for clarification (e.g., hold up an object while saying 'This?') and repair misfires (e.g., 'No, that's the spatula.') [9].

Contextual

In the example interaction, the physical context is the kitchen, a location spatially and temporally associated with particular activities (cooking, eating) and objects (utensils, dish towels, food). The child has repeatedly observed people using utensils to act upon food, making these objects more likely to be referents [10]. Indeed, objects and language are not randomly distributed across contexts but occur in highly predictable ways [10–12].

Social

Common ground. Common ground includes the tracked information, experiences, and mental states understood to be shared among interacting individuals [13]. In our case, the child's understanding of what they are doing with their parent and how their parent will react to different responses depends on how the child understands their joint activity. If the child understands the parent as giving the utensils to them, the request for the ladle is unexpected. If the child understands the activity as one of exchange, they may be ready to hand any object back. And via jointly construed relevance, some meanings (hiding, ladle) can be made extra salient [13–15]. Recent work has focused on the developmental origins of common ground cognition [16,17], including which conditions lead children to assume something is part of common ground [18,19] and how children use common ground in interaction [20–22].

Nonverbal cues. Gestures, eye gaze, facial expressions, body posture, and body position can guide interactants' attention toward a speaker's intended meaning. In our example, the parent's gaze and gesture narrow down the unknown word's referent. Such cues are also critical for establishing joint attention—an intersubjectively privileged context for learning [23–25]. Recent work has examined which factors might explain individual differences in the use of gaze and pointing [26,27].

Cognitive

Syntactic bootstrapping. The syntactic structure in which the unknown words are embedded also provides clues to their meaning. In English, the pronoun 'you' usually precedes a verb but not a noun, suggesting that 'hide' refers to an action. The demonstrative 'that' tends to come before a noun, suggesting that 'ladle' refers to an object. This process relies on linguistic knowledge that children have already acquired through other means. Syntactic bootstrapping has long figured prominently in theories of language acquisition. It is thought to be particularly important for learning verbs, adjectives, and adverbs [28–30]. Current research focuses on the developmental origins of syntactic bootstrapping [31–34] and how it interacts with other learning mechanisms [35].

Whole object assumption. The word 'ladle' could, in principle, describe different aspects of the physical object (e.g., its handle) or a higher-order category ('utensils'). The whole object assumption states that, in the absence of contradictory evidence, children expect words (nouns) to refer to entire objects at basic category levels [36]. This substantially narrows the

space of potential referents, especially when infants have little other (linguistic) information to rely on. This assumption is central to recent work testing the behavior and limits of word learning in infancy [37,38].

Cross-situational word learning. In our cooking scenario, the child has likely heard the words 'ladle' and 'hide' before. By tracking the co-occurrences of words with objects and actions across scenes, the child accumulates statistical evidence for word-meaning mappings [39]. As such, cross-situational word learning is a special case of statistical learning, which plays an important role in several aspects of language learning [40]. While evidence for cross-situational word learning has been found in experimental work [41], more recent studies examine the realization of this mechanism in naturalistic settings [11,42].

Inference via contrast. Children can use what they know about language to contrastively infer meaning in context. Like adults, children expect that different forms imply different meanings [43]. Word selection does more than identify a referent ('this shirt')—for example, expressing stance ('this rag') and specifying subdomains ('this turtleneck'). In the context of word learning, a novel word would not refer to the same exact referent as a known word. Indeed, children consistently infer that novel words map onto hitherto unnamed objects. This specific inference type is commonly referred to as 'mutual exclusivity' [44]. In our example, the child can use such inference processes to rule out meanings for 'ladle'. For example, if they have acquired the word 'spoon', they will know the referent cannot be the spoon object. Thus, the more words the child has already stored in their lexicon, the more successfully they can draw these inferences [45,46]. Inference via contrast has been an integral part of word learning theory for more than 50 years [47], and recent work has focused on its ontogenetic origins [48] and its integration with other information sources [49].

Variability in language learning environments

There is no question that children's language environments vary. In our ladle-hiding example, the child could be imagined in a wide variety of developmental contexts, each changing our assumptions about who is present, what is being cooked, whether and where the child is put down to play, and the kinds of verbal exchanges an adult is likely to initiate. Children's home language experiences are influenced by cultural and linguistic practices [3,50–53], household dynamics [1,54,55], activity contexts [56–58], socioeconomic and financial contexts [59–61], modality and perceptual access [62–65], and much more (see [66] for a recent review).

Caregivers across diverse speech communities (i.e., social groups who share a common set of norms) vary in the utility and value they attribute to talking to children. While some caregivers see infant-directed talk as a direct investment in children's economic potential [67,68], others see it as a way to elicit entertaining infant behavior [69]. Yet others doubt the usefulness of talking to someone who cannot (yet) respond [51], avoid baby talk that might elicit ill-formed vocalizations [70], or find infant-directed talk disruptive to the typical flow of social interaction [71].

Beliefs about what infant-directed talk is for do not just affect how often children are directly addressed but also what is talked about—on whose interests do interactions focus? More neoliberally oriented views of development place importance on maximizing the knowledge children glean from interaction; under this view, caregivers would be explicitly encouraged to ask about and elaborate upon their children's immediate interests [68]. An alternative strategy is to let children find their own way; caregivers socialize children to accommodate their seniors [71] and to observe and comment on the social context surrounding them, rather than expecting to be at the center of social attention [3]. We are just beginning to understand how these

intersecting and interrelated properties of children's environments influence their linguistic input. The key issue is not whether developmental environments vary (they do, tremendously) but how this variation shapes what children learn to do with language (Box 1).

Embracing variability in language learning

From a biological perspective, the particular learning environment a child will encounter is not predictable. Thus, the cognitive system supporting language learning must be adaptable. Two key implications follow: first, children successfully integrate multiple information sources to identify socially consequential meanings/functions and relate them to observed linguistic structures, and second, children fine-tune their reliance on different information sources in accordance with their specific environment. Importantly, in our view, this fine-tuning reflects true developmental adaptations in that it leads to stable individual characteristics (traits). Growing up in a particular environment impacts not just what children learn but also how they learn it.

Integrating multiple information sources

It is now commonly acknowledged that adults and children integrate multiple information sources during communication and language learning [80]—in fact, parents most likely do the same when interpreting their children's utterances [81]. However, there is still the difficult matter of specifying how such integration is done. Verbal theoretical descriptions are often vague and untestable. In this regard, computational models are appealing; they require specific, explicit assertions regarding the input, output, and process for the phenomena in focus. However, most existing models of information integration for word learning are difficult to generalize because they focus on a maximum of two information sources or on different aspects of one particular information source (e.g., moment-to-moment looking dynamics and cross-trial learning [82]). For example, one such model [83] proposed that social cues such as pointing and gaze act as a spotlight that reduces the space of potential referents and thus facilitate associative word-referent mappings. How additional information sources would play into this model is unclear.

Box 1. What type of input matters?

There has been a great deal of ink spilled over what linguistic input 'matters' for language learners—that is, what kinds of experiences with language lead to language learning. There have been at least two distinct approaches to this question in the literature. The first approach is more applied, focusing on a definition of healthy development that aligns with the goals of national economies and international institutions promoting economic development. Under this view, language development is often directly (or indirectly) emphasized for its educational (and thus economic) value rather than its cultural or communicative value. Such views emphasize findings associating more directed, interactive linguistic input with greater educational outcomes in child development (e.g., [72,73]), though short-term intervention effects suggest that this association is delicate [74,75]. The second approach, taking inspiration from human evolutionary perspectives, uses patterns of language exposure to understand what makes language learning possible, focusing on theoretical models of development. For example, recent work comparing language input across five societies finds low day-long rates of addressed talk to children across the board [1], though variation may be visible within specific activities [76]. What combination of learning mechanisms allows children to learn language from a few minutes per hour of directed speech? Characterizing exactly how linguistic information accumulates—across age, representation types, and developmental contexts—is a major puzzle to be addressed.

Thus, there is no simple answer to the question of what linguistic input 'matters' for learning language. For example, insofar as linguistic representations are concerned, overhearable and directed speech may play complementary roles in giving children the ample statistical evidence and referential grounding they need to start representing phonological forms and associating them with conventional meanings and syntactic properties [52,77–79]. And, beyond basic linguistic representations, children's input also socializes them as language users: children must figure out what to attend to during interaction, what is expected of them, what to talk about (and how to talk about it), etc. Here again, overhearable and directed speech may play complementary roles in giving children direct feedback while also showing them how interaction is negotiated in other social roles (e.g., to or by older peers). Both types of input help children track and interpret common ground across more and less familiar contexts.

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Inspired by probabilistic models of pragmatic reasoning in the Rational Speech Act tradition [84–86] and verbal theories of information integration during language learning [80,87,88], a general framework [89] has been recently proposed, according to which multiple information sources can be integrated via Bayesian inference. Information sources are either categorized as being part of an utterance (feeding into the likelihood) or the context (feeding into the prior). Importantly, this framework is open to all kinds of information sources. Each information source is thought to generate a skewed distribution over potential referents, which are then integrated into different parts of the model (Figure 2). The varying shapes and skewness of these distributions depend on an individual's sensitivity to the sources. Adaptation is the process captured by developmental changes in sensitivity; the process of information integration itself is thought to be stable across development and context. Despite its obvious benefits, the model does not provide a detailed account of how each potential information source generates its distribution. Its strength lies in providing an intuitive starting point for conceptualizing the integration process at a superordinate level. As such, it paves the way for studying multiple learning mechanisms in conjunction and helps to connect largely separate research traditions in language acquisition (Box 2).

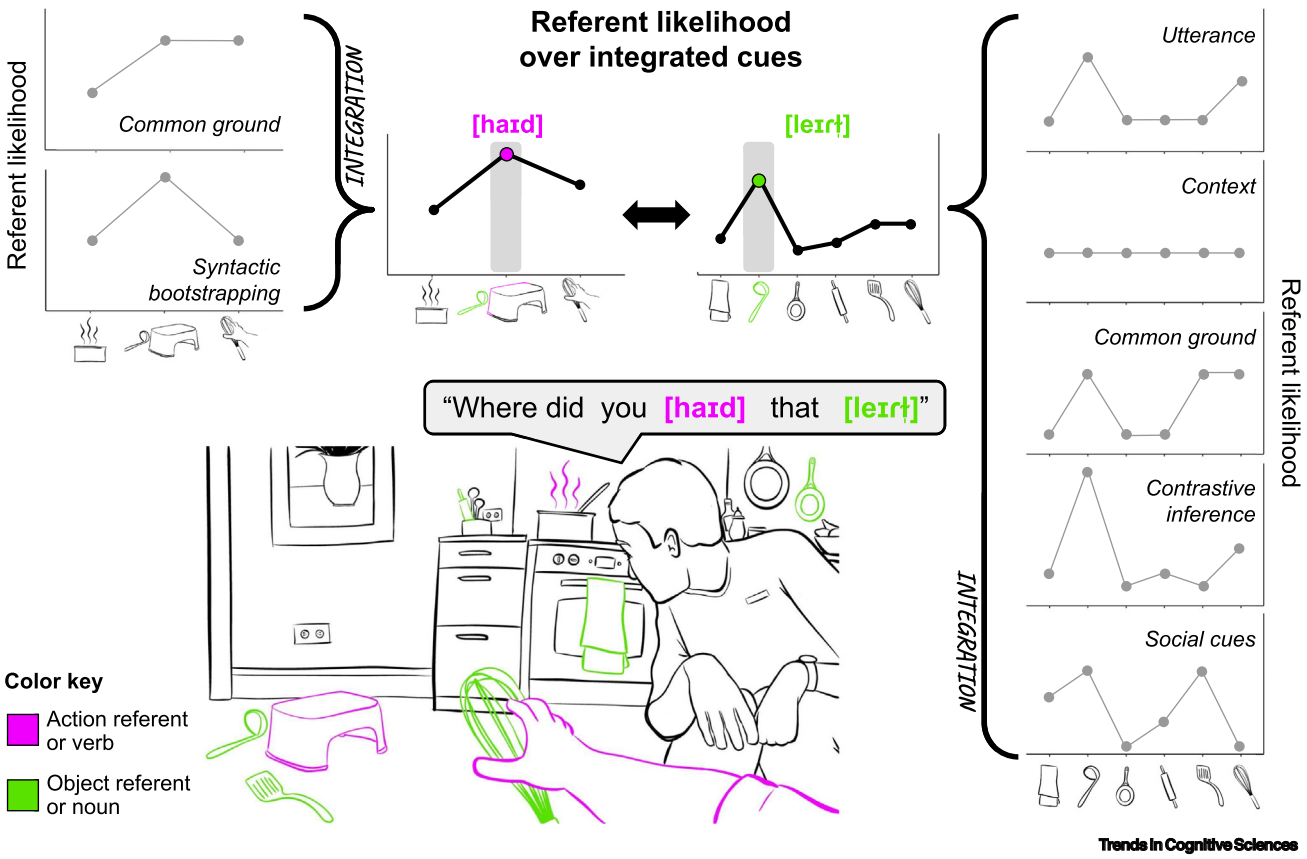


Figure 2. Schematic for integrating information from multiple mechanisms to infer meaning. In the process of word learning, children can integrate multiple information sources to infer the meanings of linguistic cues and utterances, as well as cognitive mechanisms. Each information source is assumed to generate a skewed probability distribution over potential meanings, which are subsequently integrated. Furthermore, inferences about different words are taken to be mutually constrained. Line graphs on the left (verb) and the right (noun) show the probability of each referent, given one information source. The central line graphs show the relative probability of a referent after integration over all information sources. Green (object referent) and magenta (action referent) filled dots show the most likely/inferred meaning.

Box 2. Connecting the dots

The main goal of this article is to lay out a framework that encourages researchers to jointly study what are otherwise separate aspects of language learning, including dense and naturalistic studies of everyday experiences, measures of individual differences in cognitive abilities, computational modeling, and controlled experiments. The modeling framework presented in the main text plays an important role in this process by connecting the multiple mechanisms involved. Let us spell out this research agenda. Naturalistic studies of children's everyday experiences characterize the range of similarities and differences in language learning environments. For example, by using daylong recordings, researchers can quantify the social diversity of linguistic input (e.g., how many different speakers the child encounters) as well as its linguistic properties (e.g., the number of nouns relative to other word classes). In this framework, we can generate and test predictions about how sensitive children should be to different information sources that can be gleaned from the input (reflecting adaptation). For example, children confronted with more nouns in their input should be more likely to learn nouns using inference via contrast than to learn verbs (and vice versa for children with more verbs in the input). Likewise, children confronted with a greater variety of speakers might be more sensitive to person-specific common ground. The computational framework outlined in the main text is a key part of theorizing about this adaptation because it allows us to explain how the different mechanisms and variable sensitivities appear to work together to produce language learning. The modeling results can be used to generate a diverse range of quantitative predictions that can subsequently be tested with children, e.g., cross-cultural comparisons of children's behavior in experimental conditions involving multiple information sources or the shape of developmental trajectories of vocabulary growth. In sum, this framework allows us to grapple with the complexity of language learning under real-world conditions by channeling the strengths of a broad range of approaches rooted in different disciplines.

The model and its underlying assumptions have been empirically tested in the context of word learning: after quantifying participants' sensitivity to different information sources (here: social cues, mutual exclusivity, and common ground) in separate experiments, the model was used to predict how information would be integrated and how children should act in a situation in which all information sources are available at once. Predictions were compared against actual behavioral data and were found to provide an excellent absolute and relative fit: 'absolute' in that the model explained more than 90% of the variance in the behavioral data and 'relative' in the sense that its predictions were better than alternative models. Importantly, one alternative model proposed that children selectively ignore one or more information sources in the presence of others—this model did not explain the data as well. Thus, children (and adults) appeared to use and integrate all available information sources, at the group and individual levels [46,49,90]. Taken together, the model provides a plausible and tested account of how information sources are integrated and thereby supports our first premise.

Despite the benefits of this framework, exploring the strengths and commonalities with other modeling frameworks would be a valuable way forward. Such methodological diversity is likely to produce more generalizable and comprehensive theories [91]. For example, work using this framework thus far has focused on one subprocess of language learning: identifying referents via information integration. Word learning—indeed language learning more generally—requires much more; for example, engaging long-term memory processes that are well captured by work outside what we cite (e.g., connectionist or neural network models) [92]. Another fruitful avenue is the counterintuitive result from modeling that inferring multiple things from the same data may be easier, not harder: learning words and grammar jointly allows these processes to constrain each other so that inferences about the intended meanings of utterances based on the grammatical structure can constrain word meanings, and vice versa [93,94]. Combining these approaches constitutes an important step to be taken in the future.

Adapting to the availability of information sources

The second premise, that children adapt to the availability of different information sources in their environment, has been observed but not yet explained in detail as a developmental process. Evidence for adaptation would come in the form of systematic covariation between features of

children's language learning environments and their diverging, learned sensitivities to different information sources.

Arguably, such adaptation to information sources has already been documented in the literature on children's word learning biases, though most work lacks the ecological depth that is likely necessary for pinpointing the experiential factors that underlie cross-group and cross-context adaptations. Word learning biases are sometimes treated as universal, though increasing evidence points to the contrary (see the following paragraphs), implying a substantial impact of the language learning environment on how children learn to constrain potential word meanings. We illustrate this point by briefly examining the evidence for developmental variability in two such learning biases: the noun bias and the mutual exclusivity effect.

The 'noun bias' refers to the long-observed tendency for children's first words to refer to concrete (often animate) objects [95], relating to the whole-object assumption described above [36]. Gentner [95] reasons that, because of their perceptual stability and salience, some referents (e.g., ladles) naturally evoke clearer and more consistent conceptual categories than others (e.g., hide). The naturally evoked categories are then hypothesized to appear earlier in children's vocabularies and to be more consistently circumscribed across human languages [95].

While a noun bias has indeed been observed in early word learning across typologically diverse languages, the size of the bias ranges from nonexistent to very strong across different speech communities, suggesting clear adaptation of this bias in word learning [96–100]. Children learning different languages may also weigh syntactic and pragmatic information differently in learning novel words [101]. Even within speech communities, the size of the bias depends on the observed activity context and the methods of measurement ([6]; see [102] for a systematic review), suggesting context-sensitive recalibration of information sources. Especially interesting are cases where a noun bias is supplanted by a bias for other word types, such as verbs [96,97,100].

The emergence of this conceptual bias—the noun bias—is generally tied to the child's language environment, suggesting that it derives from an adaptation to certain linguistic and/or interactional features of children's upbringing (e.g., noun ellipsis, verb position, use of imperatives, pedagogical practices). We do not yet know which experiential factors drive this process of adaptation; for example, what leads children to attend to words for 'actions' more than words for objects. But, to find our answer, we must engage with more diverse developmental contexts; more than 85% of the noun bias participants in prior work have come from just eight urban linguistic populations [102].

As described above, according to the 'mutual exclusivity' effect, children assume that novel labels refer to unknown referents [44]. Evidence from multilingual children makes it clear that this effect is driven by children's early experiences with language. Consider again our ladle-hiding example: a Spanish-English bilingual child may know the name for the ladle in Spanish ('cucharón'), but not yet in English, and so could reasonably link the word 'ladle' to the ladle *or* any other competing, non-named object (e.g., the spatula). Therefore, one can predict that multilinguals show weaker mutual exclusivity effects.

Indeed, multilingual toddlers may have weaker mutual exclusivity effects than monolingual toddlers [103–105], again suggesting clear adaptation of information sources. While both groups of children can be pragmatically shifted in their inferences about novel words, bilingual children appear to maintain a degree of flexibility that monolingual children do not [106]. The inferential

process supporting mutual exclusivity depends entirely on learned information—words that the child has already acquired and also general experience with (in)consistent word-referent mappings [45]. Therefore, the mutual exclusivity effect is a prime candidate for adaptation across developmental contexts. Multilingual populations are diverse in the relative quantities and sources of linguistic input [107]. Thus, one way future work could better understand how adaptation arises from environmental variation would be to closely examine how the emergence of the mutual exclusivity effect in children at the individual level relates to specific aspects of their early vocabulary development and lexical input (e.g., balance between languages, inventory size/diversity).

We focused on word learning biases above, but group-specific adaptations in children's use of information sources extend to other aspects of language learning; for example, how and when children attend to ambient linguistic signals. Researchers working with Mayan families have long proposed that children can be socialized to attend more closely to third-party activities, compared to North American children [50,52,77,78], though even the latter group stands to learn words through overhearing [108–110].

Even within populations, children may adapt their expectations about what kinds of speech to attend to, given the immediate context: evidence from North American and European populations shows that infants track how reliable different speakers and information sources are, adjusting their learning accordingly [111,112]. While young German children typically prefer to follow adult over peer communicative cues, presumably because adults are (on average) more informative [113], evidence of ignorance and the child relevance of the topic can flip this tendency (e.g., [114]).

In sum, beyond the lexicon, children adapt to the characteristics of their language environments in terms of when and how they attend to the myriad linguistic signals that they commonly encounter (see also [115]). To discover how these patterns in attention to input adaptively arise, we need denser, more ecologically valid coverage of more varied child communicative environments, paired with models that can help reason about children's multiple interactional threads and the integration of the multiple information sources available to them (Box 2).

Taken together, the main issue is not whether adaptation in learning mechanisms occurs, but how such adaptation of information sources relates to children's multi-threaded engagement with language. For example, how does simultaneous growth in bilingual children's vocabulary and common ground skills relate to their real-time predictions of upcoming words during a mutual exclusivity trial—for example, when they have prior experience talking to the experimenter in one language or another? We must consider not only how information sources interrelate (e.g., [116]) but how they are consistently relied upon to help solve problems, big and small, in everyday interaction. Any solution to this puzzle will likely require us to make more rigorous links between children's performance in laboratory tasks and detailed, empirical observations of their language use and input (e.g., [117]).

Moving forward

The proposal we have laid out necessitates a shift in how we approach research on language learning. Rather than isolating individual mechanisms and examining their respective developmental timelines, we argue for a more integrative perspective—one that investigates how these mechanisms interact dynamically across development. This shift requires both empirical and computational advancements to capture the complexity of language learning in a more comprehensive manner.

Our first prediction is that children rely on a multitude of available information sources. To better understand how different sources of information come together in the learning process, future research should prioritize studying learning mechanisms jointly, rather than in isolation. While past work on word learning has often focused on distinct cognitive processes, recent work has more often attempted to study how they interact [18,46,49,83]. However, developing more sophisticated models (and paradigms) that explicitly examine and test interactions between mechanisms will be crucial for advancing our understanding of information integration (see Outstanding questions).

Our second central prediction is that children's sensitivity to different information sources should adapt to the availability of these sources in their language learning environment. We have reviewed tentative evidence for this hypothesis, but establishing it as a cornerstone of language learning calls for systematic and ecologically rich research across diverse speech communities to show how environmental variation influences learning strategies [118]. For example, one might predict that children receiving substantial directed linguistic input—accompanied by ostensive signals such as eye contact and distinctive, variable pitch [119]—rely heavily on social cues for learning, whereas children with less ostensively marked input are better able to take advantage of contextual information sources (see also, e.g., [50,52,77]). This essential prediction could be tested by longitudinally tracking individuals across environments with substantial differences along these lines.

The framework we propose also has broader implications in other domains. For example, social learning and coordination have a similar multi-threaded, multi-mechanism structure and thus likely recruit the same cognitive machinery [120,121]. Another example can be found in applied domains, for instance, in human–machine interaction or chatbots based on large language models. The development of multi-threaded, multi-mechanism models advances us toward systems that can efficiently process diverse information sources in real-world communication contexts [122,123].

Outstanding questions

How can we measure children's sensitivity to different information sources across ages and contexts? How can we strike a balance between measuring the same concept across settings while doing justice to the everyday conditions for learning in a given setting?

What are the shared and distinct processes for learning different aspects of language? How do these processes constrain one another?

How can we use and compare multiple modeling frameworks to shed more light on children's use of multiple information sources while learning language?

Are children's adaptations to information sources best described as trait-level changes to their psychological processing, or are they better captured as state-level changes? What is the relation between the two?

Concluding remarks

Children learn language within the context of social interaction, driving them to solve multi-threaded problems of coordination and communication using multi-mechanism means. This feat is accomplished via children's cumulative experience with language and communication, which fine-tunes their sensitivity to information during learning. The increasing recognition of the diversity of learning environments and the availability of new tools to study integration processes have the potential to move us closer toward an understanding of variation in language learning, both in process and outcomes.

Acknowledgments

The authors thank Claire Bergey, Michael C. Frank, and Benjamin Morris for their comments on an earlier version of the paper. Manuel Bohn was funded by a Jacobs Foundation Research Fellowship (Grant No. 2022-1484-00) and a Lower Saxony Impulse Professorship through the zukunft.niedersachsen program. Marisa Casillas's work was supported by a National Science Foundation CAREER award (2238609).

References

1. Bunce, J. *et al.* (2024) A cross-linguistic examination of young children's everyday language experiences. *J. Child Lang.* 52, 786–814
2. Blasi, D.E. *et al.* (2022) Over-reliance on English hinders cognitive science. *Trends Cogn. Sci.* 26, 1153–1170
3. Ochs, E. and Schieffelin, B.B. (1984) Language acquisition and socialization: three developmental stories. In *Culture Theory: Mind, Self, and Emotion* (Shweder, R. and LeVine, R.A., eds), pp. 276–320, Cambridge University Press
4. Cychosz, M. *et al.* (2021) Vocal development in a large-scale crosslinguistic corpus. *Dev. Sci.* 24, e13090
5. Casillas, M. *et al.* (2021) Early language experience in a Papuan community. *J. Child Lang.* 48, 792–814
6. Frank, M.C. *et al.* (2021) *Variability and Consistency in Early Language Learning: The Wordbank Project*, MIT Press
7. Imai, M. and Kita, S. (2014) The sound symbolism bootstrapping hypothesis for language acquisition and

language evolution. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 369, 20130298
8. Fort, M. *et al.* (2018) Symbouki: a meta-analysis on the emergence of sound symbolism in early language acquisition. *Dev. Sci.* 21, e12659
9. Dingemanse, M. and Enfield, N. (2024) Interactive repair and the foundations of language. *Trends Cogn. Sci.* 28, 30–42
10. Roy, B.C. *et al.* (2015) Predicting the birth of a spoken word. *Proc. Natl. Acad. Sci. U. S. A.* 112, 12663–12668
11. Clerkin, E.M. and Smith, L.B. (2022) Real-world statistics at two timescales and a mechanism for infant learning of object names. *Proc. Natl. Acad. Sci. U. S. A.* 119, e2123239119
12. Tamis-LeMonda, C.S. and Masek, L.R. (2023) Embodied and embedded learning: child, caregiver, and context. *Curr. Dir. Psychol. Sci.* 32, 369–378
13. Clark, H.H. (1996) *Using Language*, Cambridge University Press
14. Clark, E.V. (2015) Common ground. In *The Handbook of Language Emergence* (Vol. 87) (MacWhinney, B. and O'Grady, W., eds), pp. 328–353, John Wiley & Sons
15. Bohn, M. and Köymen, B. (2018) Common ground and development. *Child Dev. Perspect.* 12, 104–108
16. Khu, M. *et al.* (2020) Preschoolers flexibly shift between speakers' perspectives during real-time language comprehension. *Child Dev.* 91, e619–e634
17. Bohn, M. *et al.* (2018) The social-cognitive basis of infants' reference to absent entities. *Cognition* 177, 41–48
18. Bleijlevens, N. *et al.* (2023) Pragmatics aid referent disambiguation and word learning in young children and adults. *Dev. Sci.* 26, e13363
19. Bohn, M. *et al.* (2021) Children's interpretation of ambiguous pronouns based on prior discourse. *Dev. Sci.* 24, e13049
20. Köymen, B. *et al.* (2016) Preschoolers use common ground in their justificatory reasoning with peers. *Dev. Psychol.* 52, 423
21. Siposova, B. *et al.* (2021) Common knowledge that help is needed increases helping behavior in children. *J. Exp. Child Psychol.* 201, 104973
22. Aboody, R. *et al.* (2022) Preschoolers decide who is knowledgeable, who to inform, and who to trust via a causal understanding of how knowledge relates to action. *Cognition* 228, 105212
23. Çetinçelik, M. *et al.* (2021) Do the eyes have it? A systematic review on the role of eye gaze in infant language development. *Front. Psychol.* 11, 589096
24. Kirk, E. *et al.* (2022) The relationship between infant pointing and language development: a meta-analytic review. *Dev. Rev.* 64, 101023
25. Lee, C. and Lew-Williams, C. (2023) The dynamic functions of social cues during children's word learning. *Infant Child Dev.* 32, e2372
26. Colonnesi, C. *et al.* (2024) Not too shy to point! Exploring the relationship between shyness and pointing in the second year. *Infancy* 29, 693–712
27. Ger, E. *et al.* (2023) Correlates of infant pointing frequency in the first year. *Infancy* 28, 986–1006
28. Babineau, M. *et al.* (2024) Syntactic bootstrapping as a mechanism for language learning. *Nat. Rev. Psychol.* 3, 463–474
29. Cao, A. and Lewis, M. (2022) Quantifying the syntactic bootstrapping effect in verb learning: a meta-analytic synthesis. *Dev. Sci.* 25, e13176
30. Hacquard, V. and Lidz, J. (2022) On the acquisition of attitude verbs. *Annu. Rev. Linguist.* 8, 193–212
31. Babineau, M. and Christophe, A. (2022) Preverbal infants' sensitivity to grammatical dependencies. *Infancy* 27, 648–662
32. Babineau, M. *et al.* (2023) Learning to predict and predicting to learn: before and beyond the syntactic bootstrapper. *Lang. Acquis.* 30, 337–360
33. Barbir, M. *et al.* (2023) Rapid infant learning of syntactic–semantic links. *Proc. Natl. Acad. Sci. U. S. A.* 120, e2209153119
34. Fisher, C. *et al.* (2020) The developmental origins of syntactic bootstrapping. *Top. Cogn. Sci.* 12, 48–77
35. Hacquard, V. (2023) Being pragmatic about syntactic bootstrapping. *J. Child Lang.* 50, 1041–1064
36. Markman, E.M. (1991) The whole-object, taxonomic, and mutual exclusivity assumptions as initial constraints on word meanings. In *Perspectives on Language and Thought: Interrelations in Development* (Gelman, S.A. and Byrnes, J.P., eds), pp. 72–106, Cambridge University Press
37. Campbell, J. and Hall, D.G. (2022) The scope of infants' early object word extensions. *Cognition* 228, 105210
38. Saksida, A. and Langus, A. (2024) Object labeling and disambiguation in 4-month-old infants. *Child Dev.* 95, 462–480
39. Smith, L. and Yu, C. (2008) Infants rapidly learn word-referent mappings via cross-situational statistics. *Cognition* 106, 1558–1568
40. Isbilen, E.S. and Christiansen, M.H. (2022) Statistical learning of language: a meta-analysis into 25 years of research. *Cogn. Sci.* 46, e13198
41. Roembke, T.C. *et al.* (2023) What have we learned from 15 years of research on cross-situational word learning? A focused review. *Front. Psychol.* 14, 1175272
42. Suarez-Rivera, C. *et al.* (2022) From play to language: infants' actions on objects cascade to word learning. *Lang. Learn.* 72, 1092–1127
43. Clark, E.V. (1988) On the logic of contrast. *J. Child Lang.* 15, 317–335
44. Markman, E.M. and Wachtel, G.F. (1988) Children's use of mutual exclusivity to constrain the meanings of words. *Cogn. Psychol.* 20, 121–157
45. Lewis, M. *et al.* (2020) The role of developmental change and linguistic experience in the mutual exclusivity effect. *Cognition* 198, 104191
46. Bohn, M. *et al.* (2022) Modeling individual differences in children's information integration during pragmatic word learning. *Open Mind* 6, 311–326
47. Carey, S. and Bartlett, E. (1978) Acquiring a single new word. *Pap. Rep. Child Lang. Dev.* 15, 17–29
48. Pomiechowska, B. *et al.* (2021) Twelve-month-olds disambiguate new words using mutual-exclusivity inferences. *Cognition* 213, 104691
49. Bohn, M. *et al.* (2021) How young children integrate information sources to infer the meaning of words. *Nat. Hum. Behav.* 5, 1046–1054
50. de León, L. (2011) Language socialization and multiparty participation frameworks. In *The Handbook of Language Socialization* (Duranti, A. *et al.*, eds), pp. 81–111, John Wiley & Sons
51. Gaskins, S. (2006) Cultural perspectives on infant-caregiver interaction. In *Roots of Human Sociality: Culture, Cognition and Interaction* (Enfield, N.J. and Levinson, S.C., eds), pp. 279–298, Berg
52. Rogoff, B. *et al.* (2003) Firsthand learning through intent participation. *Annu. Rev. Psychol.* 54, 175–203
53. Tamis-LeMonda, C.S. *et al.* (2012) Ethnic differences in mother–infant language and gestural communications are associated with specific skills in infants. *Dev. Sci.* 15, 384–397
54. Havron, N. *et al.* (2019) The effect of older siblings on language development as a function of age difference and sex. *Psychol. Sci.* 30, 1333–1343
55. Zukow-Goldring, P. (2002) Sibling caregiving. In *Handbook of Parenting: Being and Becoming a Parent* (Vol. 3) (Bornstein, M.H., ed.), pp. 253–286, Lawrence Erlbaum Associates
56. Soderstrom, M. and Wittebolle, K. (2013) When do caregivers talk? The influences of activity and time of day on caregiver speech and child vocalizations in two childcare environments. *PLoS One* 8, e80646
57. Tamis-LeMonda, C.S. *et al.* (2019) Routine language: speech directed to infants during home activities. *Child Dev.* 90, 2135–2152
58. Tardif, T. *et al.* (1999) Putting the "noun bias" in context: a comparison of English and Mandarin. *Child Dev.* 70, 620–635
59. Cristia, A. (2023) A systematic review suggests marked differences in the prevalence of infant-directed vocalization across groups of populations. *Dev. Sci.* 26, e13265
60. Ellwood-Lowe, M. *et al.* (2022) What causes the word gap? Financial concerns may systematically suppress child-directed speech. *Dev. Sci.* 25, e13151
61. Hoff-Ginsberg, E. (1991) Mother-child conversation in different social classes and communicative settings. *Child Dev.* 62, 782–796
62. Campbell, E.E. *et al.* (2025) Comparing language input in homes of blind and sighted children: insights from daylong recordings. *Lang. Dev. Res.* 5, 33–71

63. Goldin-Meadow, S. and Mylander, C. (1984) Gestural communication in deaf children: the effects and noneffects of parental input on early language development. *Monogr. Soc. Res. Child Dev.* 49, 782–796
64. Lu, J.C. *et al.* (2024) Becoming protactile: interactional foundations of protactile language development and language emergence. *Languages* 9, 282
65. Masataka, N. (1992) Motherese in a signed language. *Infant Behav. Dev.* 15, 453–460
66. Rowe, M.L. and Weisleder, A. (2020) Language development in context. *Annu. Rev. Dev. Psychol.* 2, 201–223
67. Lareau, A. (2011) *Unequal Childhoods: Class, Race, and Family Life*, University of California Press
68. Ochs, E. and Kremer-Sadlik, T. (2020) Ethical blind spots in ethnographic and developmental approaches to the language gap debate. *Lang. Soc.* 170, 39–67
69. Brown, P. and Casillas, M. (2025) Childrearing through social interaction on Rossel Island, PNG. In *Building Social Worlds: Thinking Forwards with Esther Newcomb Goody* (Fentiman, A.J. and Goody, M., eds), pp. 199–221, Berghahn Books
70. Schieffelin, B.B. (1983) Talking like birds: sound play in a cultural perspective. In *Acquiring Conversational Competence* (Ochs, E. and Schieffelin, B.B., eds), pp. 177–184, Routledge
71. Ochs, E. (1994) *Culture and Language Development: Language Acquisition and Language Socialization in a Samoan Village*, Cambridge University Press
72. Hart, B. and Risley, T.R. (1995) *Meaningful Differences in the Everyday Experience of Young American Children*, Paul H. Brookes Publishing
73. Hoff, E. (2003) The specificity of environmental influence: socioeconomic status affects early vocabulary development via maternal speech. *Child Dev.* 74, 1368–1378
74. Suskind, D.L. *et al.* (2016) A parent-directed language intervention for children of low socioeconomic status: a randomized controlled pilot study. *J. Child Lang.* 43, 366–406
75. Zhang, Y. *et al.* (2015) Effects of quantitative linguistic feedback to caregivers of young children: a pilot study in China. *Commun. Disord. Q.* 37, 16–24
76. Bohn, M. *et al.* (2024) Mealtime conversations between parents and their 2-year-old children in five cultural contexts. *Dev. Psychol.* 60, 1255–1268
77. Foushee, R. and Srinivasan, M. (2024) Infants who are rarely spoken to nevertheless understand many words. *Proc. Natl. Acad. Sci. U. S. A.* 121, e2311425121
78. Shneidman, L.A. *et al.* (2009) Social experience, social attention and word learning in an overhearing paradigm. *Lang. Learn. Dev.* 5, 266–281
79. Cristia, A. (2020) Language input and outcome variation as a test of theory plausibility: the case of early phonological acquisition. *Dev. Rev.* 57, 100914
80. Tomasello, M. (2019) *Becoming Human: A Theory of Ontogeny*, Harvard University Press
81. Meylan, S.C. *et al.* (2023) How adults understand what young children say. *Nat. Hum. Behav.* 7, 2111–2125
82. Bhat, A.A. *et al.* (2022) Word-object learning via visual exploration in space (WOLVES): a neural process model of cross-situational word learning. *Psychol. Rev.* 129, 640
83. Yu, C. and Ballard, D.H. (2007) A unified model of early word learning: integrating statistical and social cues. *Neurocomputing* 70, 2149–2165
84. Degen, J. (2023) The rational speech act framework. *Annu. Rev. Linguist.* 9, 519–540
85. Franke, M. and Jäger, G. (2016) Probabilistic pragmatics, or why Bayes' rule is probably important for pragmatics. *Z. Sprachwiss.* 35, 3–44
86. Frank, M.C. and Goodman, N.D. (2012) Predicting pragmatic reasoning in language games. *Science* 336, 998
87. Hirsh-Pasek, K. (2000) An emergentist coalition model for word learning: mapping words to objects is a product of the interaction of multiple cues. In *Becoming A Word Learner: A Debate on Lexical Acquisition* (Golinkoff, R.M. *et al.*, eds), pp. 136–164, Oxford University Press
88. Bates, E. and MacWhinney, B. (2014) Competition, variation, and language learning. In *Mechanisms of Language Acquisition* (MacWhinney, B., ed.), pp. 157–193, Psychology Press
89. Bohn, M. and Frank, M.C. (2019) The pervasive role of pragmatics in early language. *Annu. Rev. Dev. Psychol.* 1, 223–249
90. Bohn, M. *et al.* (2022) Predicting pragmatic cue integration in adults' and children's inferences about novel word meanings. *J. Exp. Psychol. Gen.* 151, 2927
91. Dale, R. *et al.* (2023) The fundamental importance of method to theory. *Nat. Rev. Psychol.* 2, 55–66
92. Wojcik, E.H. *et al.* (2022) The map trap: why and how word learning research should move beyond mapping. *Wiley Interdiscip. Rev. Cogn. Sci.* 13, e1596
93. Maurits, L. *et al.* (2009) Joint acquisition of word order and word reference. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 31)
94. van Tiel, B. *et al.* (2024) Word order and the learnability of artificial languages. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 46)
95. Gentner, D. (1982) *Why nouns are learned before verbs: linguistic relativity versus natural partitioning (technical report no. 257)*, Center for the Study of Reading, National Institute of Education
96. Casillas, M. *et al.* (2024) Little evidence for a noun bias in Tseltal spontaneous speech. *First Lang.* 44, 600–628
97. De León, L. (1999) Verbs in Tzotzil (Mayan) early syntactic development. *Int. J. Biling.* 3, 219–240
98. Gopnik, A. and Choi, S. (1990) Do linguistic differences lead to cognitive differences? A cross-linguistic study of semantic and cognitive development. *First Lang.* 10, 199–215
99. Setoh, P. *et al.* (2021) Contrasting lexical biases in bilingual English–Mandarin speech: verb-biased mothers, but noun-biased toddlers. *J. Child Lang.* 48, 1185–1208
100. Tardif, T. (1996) Nouns are not always learned before verbs: evidence from Mandarin speakers' early vocabularies. *Dev. Psychol.* 32, 492–504
101. Imai, M. *et al.* (2008) Novel noun and verb learning in Chinese-, English-, and Japanese-speaking children. *Child Dev.* 79, 979–1000
102. Zhang, Y. *et al.* (2026) The noun bias: a systematic review 1973–2024. (preprint).
103. Byers-Heinlein, K. and Werker, J.F. (2009) Monolingual, bilingual, trilingual: infants' language experience influences the development of a word-learning heuristic. *Dev. Sci.* 12, 815–823
104. Houston-Price, C. *et al.* (2010) Language experience shapes the development of the mutual exclusivity bias. *Infancy* 15, 125–150
105. Kalashnikova, M. *et al.* (2018) The development of fast-mapping and novel word retention strategies in monolingual and bilingual infants. *Dev. Sci.* 21, e12674
106. Kalashnikova, M. *et al.* (2015) The effects of linguistic experience on the flexible use of mutual exclusivity in word learning. *Biling. Lang. Cogn.* 18, 626–638
107. Unsworth, S. (2016) Quantity and quality of language input in bilingual language development. In *Bilingualism across the Lifespan: Factors Moderating Language Proficiency* (Nicoladis, E. and Montanari, S., eds), pp. 103–121, American Psychological Association
108. Fitch, A. *et al.* (2020) Toddlers' word learning through overhearing: others' attention matters. *J. Exp. Child Psychol.* 193, 104793
109. Floor, P. and Akhtar, N. (2006) Can 18-month-old infants learn words by listening in on conversations? *Infancy* 9, 327–339
110. Foushee, R. *et al.* (2021) Self-directed learning by preschoolers in a naturalistic overhearing context. *Cognition* 206, 104415
111. Koenig, M.A. and Harris, P.L. (2005) Preschoolers mistrust ignorant and inaccurate speakers. *Child Dev.* 76, 1261–1277
112. Tummeltshammer, K.S. *et al.* (2014) Infants track the reliability of potential informants. *Psychol. Sci.* 25, 1730–1738
113. Kachel, G. *et al.* (2021) Toddlers prefer adults as informants: 2- and 3-year-olds' use of and attention to pointing gestures from peer and adult partners. *Child Dev.* 92, e635–e652
114. VanderBorght, M. and Jaswal, V.K. (2009) Who knows best? Preschoolers sometimes prefer child informants over adult informants. *Infant Child Dev.* 18, 61–71
115. Wermelinger, S. *et al.* (2024) From everyday exposure to pragmatic mastery: the COME perspective. *Int. Rev. Pragmat.* 16, 149–161
116. Golinkoff, R.M. *et al.* (1994) Early object labels: the case for a developmental lexical principles framework. *J. Child Lang.* 21, 125–155

117. Bowerman, M. (1980) The structure and origin of semantic categories in the language learning child. In *Symbol as Sense* (Foster, M.L. and Brandes, S.H., eds), pp. 277–299, Academic Press
118. Casillas, M. (2023) Learning language *in vivo*. *Child Dev. Perspect.* 17, 10–17
119. Csibra, G. and Gergely, G. (2009) Natural pedagogy. *Trends Cogn. Sci.* 13, 148–153
120. Gweon, H. (2021) Inferential social learning: cognitive foundations of human social learning and teaching. *Trends Cogn. Sci.* 25, 896–910
121. De Felice, S. *et al.* (2023) Learning from others is good, with others is better: the role of social interaction in human acquisition of new knowledge. *Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci.* 378, 20210357
122. Yildirim, I. and Paul, L. (2024) From task structures to world models: what do LLMs know? *Trends Cogn. Sci.* 28, 404–415
123. Ge, Z. *et al.* (2024) WorldGPT: empowering LLM as multimodal world model. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pp. 7346–7355